After how many minutes (neares interger) with the two hands be again lying one above the other.

(a) 60 (b) 62

(c) 65 (d) 67

Ans. (c) Required time $= \frac{60}{55} \times 60 \text{ min} = 65\frac{5}{11} \approx 65$ min

57. There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. 10 student can speak both English and Hind. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi and how many can speak only English.

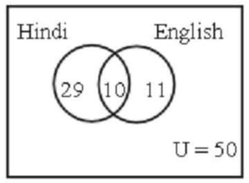
(a) 21, 11 and 29 respectively

(b) 28,18 and 22 respectively

(c) 37, 27 and 13 respectively

(d) 39, 29 and 11 respectively

Ans. (d)



39 can speak Hindi.

29 Can only speak Hindi

11 Can only speak English

58. A gardener increased the area of his rectangular garden by increasing its length by 40% and decreasing its width by 20%. The area of the new garden.

(a) has increased by 20%

(b) has increased by 12 %

(c) had increased by 8%

(d) is exactly the same as the old area

Ans. (b)

The area of the new garden

$$= \left(40 - 20 - \frac{40 \times 20}{100}\right)\%$$

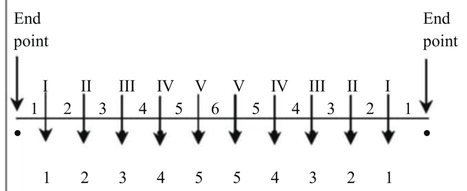
$\Rightarrow +20 - 8 = +12\% = 10\%$ Increase.

59. A straight line segment is 36 cm long. Points are to be marked on the line from both the end points. From each end the, first point is at a distance of 1 cm from the end, the second point is at a distance of 2 cm from the first point and the third point is at a distance of 3 cm from the second point and so on. If the points on the ends are not counted and the common points are counted as one, what is the number of points ?

(a) 10 (b) 12

(c) 14 (d) 16

Ans. (b) The line with points as per the data would look like this.



Number of points = 10

* According to UPSC (b).

60. If Sohan, while selling two goats at the same price, makes a profit of 10% on one goat and suffers a loss of 10% on the other

(a) he make no profit no loss

(b) he makes a profit of 1%

(c) he suffers a loss of 1%

(d) he suffers a loss of 2%

Ans. (c) Required profit/loss

$$= \left(10 - 10 - \frac{10 \times 10}{100}\right)\%$$

$= -1\%$

$\because$ Hence, loss of 1%

61. Out of a total of 120 musicians in a club, 5% can play all the three instruments, guitar violin and flute. It so happens that the number of musicians who can play any two and only two of the above instruments is 30. The number of musicians who can play the guitar alone is 40. What is the total number of those who can play violin alone of flute alone?

(a) 45 (b) 44

(c) 38 (d) 30

Ans. (b) The number of musicians who play three instruments

$= 120 \times 5\%$

$= 120 \times \frac{5}{100} = 6$

$= 6$

The number of musicians who play two instruments

$= 30$

The number of musicians who play only guitar

$= 40$

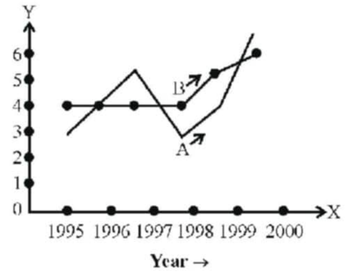
Then, the total number of those who can play violin done of flute alone

$= 120 - (6 + 30 + 40)$

$= 120 - (76) = 44$

Directions for the following 4 (four) items :

The following graph shows the average profit of two fruit-sellers A and B in thousands (R) per year from the year 1995 to 2000. Consider the graph and answer the 4 (four) items that the follow:



62. In which year is the average profit of A and B same ?

(a) 1995 (b) 1996

(c) 1997 (d) 1998

Ans. (b) The average profit of A and B same is year 1996.

63. What is the difference between the average profit of B and A in the year 1998 ?

(a) – Rs.100 (b) – Rs.1,000

(c) + Rs.600 (d) – Rs. 300

Ans. (c) The difference between the average profit of B and A in the year 1998

$= 4000 - 3400$

$= 600$

64. How much more average profit did A make in the year 2000 than in the year 1999 ?

(a) Rs. 200 (b) Rs. 1,000

(c) Rs. 1,500 (d) Rs. 2,000

Ans. (d) The trend of the average profit of B from the year 1997 to the year 2000 is non-decreasing.

65. What is the trend of the average profit of B from the year 1997 to the year 2000?

(a) Non- Increasing

(b) Non-decreasing

(c) Steady

(d) Fluctuating

Ans. (b) According to the Graph in the question, we have

The average profit of B from the year 1997 to the year 2000 is :

1997 = 4000

1998 = 4000

1999 = 5000

2000 = 5400

Hence, The Average profit of B from the year 1997 to 2000 is Non- decreasing.

66. The following table shows the marks obtained by two students in different subjects :

	Student A	Maximum Marks	Student-B	Maximum Marks
English	60	100	80	150
Psychology	70	100	70	100
History	50	100	60	100
Sanskrit	30	50	15	25

The difference in the mean aggregate percentage marks of the students is

(a) 2.5% (b) 13.75%

(c) 1. 25% (d) Zero

Ans. (d) The mean of A (in %)

$$= \left(\frac{60 + 70 + 50 + 30}{100 + 100 + 100 + 50}\right) \times 100\%$$

$$= \frac{210}{350} \times 100\%$$

$= 60\%$

The mean of B (in %)

$$= \left(\frac{80 + 70 + 60 + 15}{150 + 100 + 100 + 25}\right) \times 100\%$$

$$= \frac{225}{375} \times 100\%$$

$= 60\%$

The required difference

$= 60\% - 60\%$

$= 0$

67. The following table gives population and total income of a city for four years :

Year	1992	1993	1994	1995
Population in lakhs	20	21	22	23
Income in croes (Rs.)	1010	1111	1225	1345